JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

MECHANICAL WORKSHOP CELL & CAD/CAM CENTER

DEPARTMENT OF MECHANICAL ENGINEERING

★ OFFICIAL LABORATORY MANUAL & PRACTICE PROBLEMS ★

Computer Aided Drafting (CAD) & Engineering Graphics Lab Manual

Course Code:	R23-MEC-LAB1	**Academic Scheme:**	I B.Tech I Semester (R23 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Lab Exercise 1: CAD GUI & Coordinate Systems

Setting up drawing environment, limits, units, grid, snap, Absolute and relative polar coordinate drafting in AutoCAD.

Lab Exercise 2: Geometric Constructions & Conic Sections

Drafting ellipse, parabola, hyperbola using concentric circles and eccentricity methods, Involutes and cycloidal curves.

Lab Exercise 3: Orthographic Projections of Points & Lines

Projections of points in 4 quadrants, True length and inclination of oblique straight lines with traces.

Lab Exercise 4: Projections of Regular Solids

Prisms, pyramids, cylinders and cones inclined to one and both reference planes with sectional views.

Lab Exercise 5: 3D Modeling & Isometric Assembly

Extrude, Revolve, Sweep commands, Modeling flanged coupling, socket-spigot joint, and knuckle joint.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Day-to-day Lab Work & Record	15 Marks
Internal Lab Practical Test	15 Marks
External Practical Viva-Voce Exam	50 Marks (Total: 80 Marks)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM